Ram Treatment of the Higher-Dimensional Charged Penrose Reduction

Exact Operators, Local Certificates, and Geometric Obstructions

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Abstract

This report audits the pinned October 5, 2026 charged reduction in spatial dimensions $n \geq 4$, with its neutral spacetime Penrose theorem retained as an explicit dependency. We read all scientific modules, identify the exact rank-one nondivergence and natural-growth operators, and route the relevant Ram corpus results through their actual hypotheses. The local contributions are a rank-one strong residual certificate with a proof of its operator restrictions, a quadratic-gradient function-space obstruction, a complete first/second derivative specification for the coupled filled problem, a bounded-domain flux-conversion contraction certificate, and charge-form and graph-magnetic obstructions. Exact algebra and bounded arithmetic tests are separated from analytic proofs and from unperformed geometric computations. No independent certification of the global charged inequality, high-dimensional singular-multiplier elimination, neutral input, or global rigidity is claimed. Applicability level L4 is not a contribution score. Contribution assessment: C2/5, source-specific local certificates and obstructions with complete restricted proofs, but no certified solve of the source geometry or new global Penrose theorem.

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1 Scope, Provenance, and Evidence Contract

The assigned source is OpenAI, *A Charged Reduction of the Spacetime Penrose Inequality in Spatial Dimensions at Least Four*, October 5, 2026, at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Write `S` for its directory under `_sources/openai-math/preprints/`; its full directory name is the title with hyphens followed by `-October-5-2026`. All source labels below are literal labels in `S/build/sections/`. The read set comprises README, paper assembly, preamble, bibliography, both figures, and *all* eleven scientific modules `00-introduction.tex` through `10-recovery-sharpness.tex`. Reading a proof is not a formal verification of it. This report distinguishes (i) source assertions and their dependencies, (ii) proved local statements here, and (iii) finite computations.

The existing `_audit/upstream-20261008` finding records unchanged selected contents through `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Its selected-file and one-hop checks are reused, not recrawled. They establish version provenance, not mathematical correctness. Row 511 of `00-Applicability-Catalogue/applicability.json` supplies the L4 routing; all three `_audit/corpus-core-map.txt`, `corpus-foundations-map.txt`, and `corpus-applications-map.txt` were read as routing maps, not as proof substitutes.

The actual required companion statement was read in *The spacetime Penrose inequality and enclosing area*, September 27, 2026, `build/sections/introduction.tex`, lines 1–230, especially `def:data`, `def:fullcuts`, and `thm:main`. We import its numerical statement only as a dependency. We do not claim to have reverified its full proof, import its separate automatic-timelikeness corollary, or substitute a Riemannian theorem for it. The three-dimensional charged companion is contextual, not an additional theorem used here.

This M511 report concerns the $k = n - 1 \geq 3$ algebra, rank-one PDE, filled-source flux conversion, magnetic two-form constraint, and high-dimensional singular-set interfaces. It does not duplicate a three-dimensional or general neutral-author report. No other report or shared catalogue is modified.

2 What the Source Actually Claims

Set $k = n - 1$, $\omega = \mathrm{Area}(S^k)$ and $c_n^2 = k(k-1)/2$. The Laplacian is $\operatorname{div}\nabla$, with negative Fourier symbol. The data are smooth, oriented, connected, complete including a nonempty compact boundary, have exactly one coordinate end and compact topology off it, and are purely electric. With $\tau = \operatorname{tr}_g K$,

$$2\mu_m = R_g + \tau^2 - |K|^2 - k(k-1)|E|^2, \qquad J_m = \operatorname{div}(K - \tau g), \quad \mu_m \geq |J_m|, \quad \operatorname{div} E = 0. \tag{2.1}$$

The source assumes integrability of $\mu_m, |J_m|$, falloff $g - \delta = O_2(r^{-q})$, $K = O_1(r^{-1-q})$, $E = O_1(r^{-k})$ with $q > (n-2)/2$, and finite ADM fluxes. The energy and momentum normalizations are $1/(2k\omega)$ and $1/(k\omega)$; $Q = \omega^{-1} \int E \cdot \nu$. Every boundary component satisfies $H + \epsilon_a \operatorname{tr}_S K \leq 0$, with independently chosen $\epsilon_a \in \{-1, 1\}$.

The enclosing area a is the infimum of areas of *full* cuts: the entire intrinsic boundary of a closed connected end-containing domain. All components and contact portions count. Define

$$r_A = (a/\omega)^{1/k}, \qquad X = r_A^{k-1}, \qquad q_e = |Q|. \tag{2.2}$$

Neither a selected component nor a numerically convenient level sheet is a replacement for a full cut.

2.1 Explicit Neutral Dependency

Source `01-statements.tex`, `thm:neutral-input`, assumes the numerical neutral spacetime Penrose theorem for smooth one-ended data in every $d \geq 3$, with integrable dominant-energy densities, O_6/O_5 asymptotics at a rate $(d-2)/2 < q_0 < d-2$, and *future* weakly trapped boundary on every component. Positive enclosing area $A_* > 0$ and $E_{\mathrm{ADM}} > |P_{\mathrm{ADM}}|$ are hypotheses of that input. Its conclusion is

$$\sqrt{E_{\mathrm{ADM}}^2 - |P_{\mathrm{ADM}}|^2} \geq \frac{1}{2}(A_*/\omega_{d-1})^{(d-2)/(d-1)}. \tag{2.3}$$

Every global conclusion below remains conditional on this input and on the intervening source arguments.

2.2 Branches and Rigidity Restrictions

Source `thm:numerical` concludes $a > 0$, future timelikeness of the ADM vector, $m = \sqrt{E_{\mathrm{ADM}}^2 - |P_{\mathrm{ADM}}|^2} \geq q_e$, and

$$X \leq m + \sqrt{m^2 - q_e^2}. \tag{2.4}$$

For $X > q_e$ this gives the polynomial branch $m \geq (X + q_e^2/X)/2$. For $X \leq q_e$, the reduction gives only the mass–charge bound. It does not supply the polynomial inequality on that branch. Source `thm:rigidity` additionally requires a connected outer-area-minimizing future marginal boundary, no smooth interior full weakly future-trapped cut, $m > q_e$, and outer equality. It identifies the original data with a Reissner–Nordstrom–Tangherlini (RNT) slice with *zero induced magnetic two-form*. It is not an extremal, disconnected-boundary, or behind-the-horizon classification.

3 Step-by-Step Source Audit

The following entries identify both the mechanism read and what the Ram treatment can actually support. "Source" is attribution, not independent proof certification.

Module	Scientific step, exact locators, and audit boundary
00, 01	Introduction and theorem scope. `thm:neutral-input`, `thm:numerical`, `thm:rigidity`, `thm:converse`. Keep branch and magnetic restrictions; no representation theorem supplies the neutral input.
02 preparation	`prep:positive-area` uses a bounded closed calibration and a tube, not DEC. `prep:compact-inverses` needs constant/affine scalar moments for double divergence and translation/rotation moments for symmetric divergence. These compatibility conditions cannot be deleted by matrix inversion.
02 continued	`prep:strictification` is a mixed-boundary nonlinear scalar solve. `prep:end-increase` changes energy with fixed charge. `prep:rest-preparation` retains the electric form rather than applying an arbitrary electromagnetic boost. `prep:source-extension` permits sources only inside the filling. `prep:prepared-reduction` is the one-parameter inequality, not the final branch optimization.

Module	Scientific step, exact locators, and audit boundary
03 filled system	`alg:definitions`, `alg:implicit-differential`, `alg:tensor-comparison`, `alg:scalar-identity`, `alg:completed-squares`, `alg:system`. Exact rank-one nondivergence trace plus a divergence natural-growth equation. Our derivatives and local certificates preserve these operators. The floor exclusion and global height estimates remain source arguments.
04 elliptic	`pde:special-gradient`, `pde:face-correction`, `pde:frozen-derivative`. Gradient regularity uses gradient-aligned structure, a curved-face correction, and only then a differentiated frozen-axis divergence equation. It does not convert the original trace equation to a symmetric variational problem.
04 continued	`pde:natural-growth`, `pde:normalized-trace`, `pde:coupled-estimates`, `pde:trial-trace`, `pde:filled-solve`, `pde:end-equation`, `pde:mass-cost`, `pde:charged-region`. Morrey control feeds the second solve; degree and fixed-N exhaustion are distinct from a local inverse certificate. Polynomial mass-cost control is distinct from unrestricted fixed-N Schauder constants.
05 flux conversion	`flux:conversion-equation`, `flux:background-equation`, `flux:dimension-slack`, `flux:trace-tensor`. The flux equation uses the actual filled background tensor and retained electric form. The final neutral cut is late and full. Its plateau may be chosen after N without invalidating the gradient budget. Our smallness certificate is sufficient, not the source's general degree argument.
06 original variation	`var:area-envelope`, `var:multiplier-identity`, `var:trace-free-sheets`, `var:moment-regularity`. Area supergradients are measures on minimizing enclosures. The singular normal-normal Hessian term and actual gravitational adjoint must be retained. Finite moments alone do not show an area envelope differentiable.
07 singular multipliers	`var:atomic-leaves`, `var:common-jacobi`, `var:weak-focusing`, `var:capacity-submean`, `var:cone-subsolution`, `var:reciprocal-jacobi`, `var:diffuse-zero-mass`, `var:area-elimination`. Atomic jumps, diffuse BV locality, and the singular-set capacity argument are logically separate. Sampling cannot establish any of these global eliminations.
07 stationary data	`var:stationary-data` gives smooth lapse/shift/potential after measure elimination. The global electric potential uses all compactly supported closed test forms, not just exact ones. No prior simple connectivity is assumed.
08 stationary rigidity	`stat:base-attachment`, `stat:vacuum-reduction`. Electric-flow preservation gives alignment; logarithmic-energy arguments cross the zero set of $W = u^2 - \lvert X\rvert^2$. Positive and zero boundary lapse have different smooth boundary attachments. The vacuum reduction is not supplied by our scalar inverse.
09 vacuum uniqueness	`uniq:end-expansion`, `uniq:compact-inverse`, `uniq:correction-mass`, `uniq:smooth-nonnegative-mass`, `uniq:rough-zero-mass`, `uniq:vacuum-uniqueness`. The compact-source inverse uses $n/2 < s < n$, $p = ns/(n-s) > n$, weighted tails, and small compact $W^{2,s}$ metric changes. The nonnegative-mass step invokes the already completed numerical theorem at $Q = 0$ and therefore retains the neutral dependency.
10 recovery	`rec:original-embedding`, `rec:global-time`, `rec:ingoing`, `rec:kruskal`, `rec:sharpness`. A global time primitive is obtained only after topology is identified. Boundary regularity is checked in ingoing or Kruskal coordinates. Radial tilted slices have zero magnetic pullback and need not have $K = 0$.

4 Actual Corpus Transfers, Not Map-Level Analogies

The common root for the following three read private files is

`C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/`.

These locators are exact and the indicated proofs, not merely catalogue titles, were read.

File and read extent	Transfer and hypothesis boundary
`TR8-lin-non-lin/` `TR8-lin-non-lin.tex`, lines 81–323	`sec:elliptic`, `thm:coercive`, `eq:operator-domain`, `eq:compatibility`, `thm:green-equivalence`, `thm:heat-equivalence`, `sec:semilinear`. The elliptic proof concerns $L = -\operatorname{div}(A\nabla) + c$, Hermitian measurable uniformly positive A, $c \geq 0$, on the actual form domain H_0^1. Its inverse is $H^{-1} \to H_0^1$. The realized L^2 domain need not be H^2. Boundary lifts and Neumann compatibility are part of the theorem. Transfer to flux conversion is exact; transfer to $A_\chi : \operatorname{Hess} f$ as if it were this operator is rejected. The heat result is not needed. The cubic example's dimension restriction is not a license for natural growth in $n \geq 4$.
`TR6-lin-non-lin/` `TR6-lin-non-lin.tex`, lines 324–410 and 506–669	The theorem titled "Full Urysohn variation", `eq:du`, `eq:ddu`, `eq:newtonfull`, and `sec:shape`. First variation includes source and observation state derivatives; second variation has all four placements. This supports retaining all coefficient derivatives in Section 8, not just a frozen Green kernel. The shape proof needs regular moving boundaries, endpoint/Jacobian terms, and transverse levels; it cannot differentiate a nonsmooth infimum of areas.
`TR1-lin-nonlin/` `stability.tex`, full file	`thm:pert`, `eq:weak`, `eq:boundaryweak`, `thm:ball`, `eq:radii`, `prop:transfer`. Finite perturbation needs an actual singular-value gap exceeding operator error, with physical norms. Weak adjoints differentiate coefficient times test and retain boundary traces. The ball theorem requires a bounded isomorphism on the full Banach spaces, not a finite left inverse. Transfer requires bounded inverse maps and the full chain rule.

This is a specific Ram treatment: exact operator realization; boundary-aware weak moment assembly where legitimate; complete nonlinear differentiation; finite/infinite error separation; and feasibility-preserving residual balls. It is not a claim that a change of representation proves the geometric theorem.

5 Exact Filled Equations and Their Function Spaces

Fix the prepared background and $N > \max\{k, 2/\varepsilon\}$. Let $p(t) = N\theta(Nt)$, $l' = pl$, $l(0) = 1$, $0 \leq p \leq N$, $l = e^{Nt}$ for $t \leq 0$ and $l \leq e$. Put $\ell = e^{-N\varepsilon}$ and $\eta = \ell^{3/2}$. For $q = \nabla f$, set

$$D = 1 + l(t)^2|q|^2, \qquad Z = t + (2k)^{-1}\log D, \tag{5.1}$$

$$c = lD^{-1/2}, \qquad w = cq, \qquad v = |w|^2, \qquad d = 1 - v = D^{-1},$$

$$\chi = \frac{kd}{k+pv}, \qquad A_d = g^{-1} - w \otimes w, \qquad A_\chi = g^{-1} - \frac{k+p}{k+pv} w \otimes w. \tag{5.2}$$

The implicit derivative in t is $1 + pv/k \geq 1$, so $t = t(Z, q)$ is single-valued and smooth, including at $q = 0$. Further,

$$u = e^{(k-1)t} l\sqrt{D}, \qquad B = K + c\,\mathrm{Hess}\, f, \qquad F = A_\chi : B, \tag{5.3}$$

$$V = uA_\chi(k\mathrm{d}Z + K(w, \cdot)), \qquad k\mathrm{d}Z = (k + pv)\mathrm{d}t + c\,\mathrm{Hess}\, f(w, \cdot). \tag{5.4}$$

The filled system on $\mathcal{M}_{N,R}$ is exactly

$$F = \eta f, \qquad \mathrm{div}\, V = u\Xi, \qquad f = Z = 0 \quad \text{on the outer sphere}, \tag{5.5}$$

where

$$\Xi = \delta_0(\mathcal{T} + \eta c|q|^2) + \rho - m_0 C_N(\rho_0 + v\rho_1), \quad m_0 = \theta(N(t + \varepsilon)). \tag{5.6}$$

Here $\rho_0 \sim r^{-n-\beta}$, $\rho_1 \sim r^{-2b_1}$ and $\rho = c_*\rho_0$ with $n/4 < b_1 < \min\{n - 2, n/2\}$. Such an interval is nonempty for $n \geq 4$, and ρ_1^2 is integrable because $4b_1 > n$. Filling interfaces are *not* internal Dirichlet boundaries; no DEC is imposed in the filling.

In the slope direction A_χ has eigenvalue χ and in every transverse direction eigenvalue one. In particular

$$\frac{k}{k + N} A_d \leq A_\chi \leq A_d \leq g^{-1}. \tag{5.7}$$

The first equation is

$$A_\chi : \mathrm{Hess}\, f = \frac{\sqrt{D}}{l}(\eta f - \mathrm{tr}_{A_\chi} K). \tag{5.8}$$

It is a *nondivergence* equation. Replacing it by $\mathrm{div}(A_\chi \nabla f)$ adds a generally nonzero coefficient-derivative drift.

5.1 Natural Growth and the Correct Order of Regularity

The second equation has the source form

$$-\mathrm{div}(\mathcal{A}\nabla Z) = \mathrm{div}\, \mathcal{B} + \mathcal{F}, \quad \mathcal{A} = kuA_\chi, \quad \mathcal{B} = uA_\chi K(w, \cdot), \quad \mathcal{F} = -u\Xi. \tag{5.9}$$

On an established fixed-N bounded solution set, $|\mathcal{F}| \leq C_N(1 + |\mathrm{d}Z|^2 + |\,\mathrm{Hess}\, f|^2)$. The source first obtains gradient Holder regularity for the *gradient-aligned* first equation and the Morrey estimate $\int_{B_r} |\,\mathrm{Hess}\, f|^2 \leq C_N r^{n-2+2\alpha}$. The natural-growth lemma assumes bounded individually smooth Z, uniformly elliptic measurable $\mathcal{A}$, bounded $\mathcal{B}$, and $|\mathcal{F}| \leq a_0|\mathrm{d}Z|^2 + \mu$ with $\mu(B_r) \leq Cr^{n-2+\alpha_0}$, $0 < \alpha_0 < 1$. Exponentials $e^{\pm LZ}$ absorb the quadratic term. These hypotheses precede the Holder and Schauder bootstraps; smoothness of each approximant is not a uniform bound.

The constant-boundary gradient proof uses odd reflection of the scalar unknown, even reflection of metric and axial coefficient, and the explicit curved-face flux correction in `pde:face-correction`. Only after freezing an *axis* does differentiation give

$$\mathrm{div}\big[(I - e \otimes e + \chi e \otimes e)\nabla(\partial_e Y_0)\big] = 0. \tag{5.10}$$

This is not a divergence reformulation of (5.8) for variable $e = \nabla f/|\nabla f|$.

5.2 Boundary and Exhaustion Ledger

For the strictification scalar ϕ, the source uses $-\Delta\phi = B\sqrt{|\mathrm{d}\phi|^2 + r^{-2k}} + r^{-n-\beta}$, with normal derivative -1 at the original inner boundary (normal directed into the exterior), Dirichlet zero at the far truncated sphere, and decay at infinity. The homogeneous linearized mixed traces belong to $-\Delta h - B\mathrm{d}\phi \cdot \mathrm{d}h/\sqrt{|\mathrm{d}\phi|^2 + r^{-2k}}$. For the filled system the two unknowns instead have outer Dirichlet zero and no inner boundary. For flux conversion below, $Y = 0$ at that same artificial outer boundary and $Y \to 0$ at infinity. For a nonzero prescribed trace, one must first choose a lift in the actual energy/strong space and include its operator residual. Pure Neumann formulations additionally need the relevant flux compatibility and quotient by constants; they are not these problems.

The source solves at fixed N by compact degree, using a moving admissible region with $\min t > -\varepsilon$, and then exhausts $R \to \infty$. The asymptotic tails are $Z, t = O_j(r^{2-n})$ and exponential decay of f. Constants can grow exponentially in a polynomial of N. The mass-flux lower bound is much more selective: its adverse contribution is controlled by $P_N(\ell + \ell^2/\eta) = P_N(\ell + \ell^{1/2}) \to 0$. A numerical certificate at one N, R proves neither the floor exclusion for all exhaustions nor this mass-cost estimate. The limit order is fixed preparation, fixed s, fixed profile and ε, then $N \to \infty$, $\varepsilon \downarrow 0$, profile error to zero, removal of preparation, and finally optimization in s.

6 A Strong Certificate That Preserves Nondivergence Form

We give a bounded-domain result, not a replacement for the source's special-gradient theorem. Let U be a bounded convex C^2 Euclidean domain for which the Dirichlet Laplacian is an isomorphism $\Delta_D : H^2(U) \cap H_0^1(U) \to L^2(U)$. The result also holds on a rectangular box, by its sine expansion. Write $\mathcal{X} = H^2 \cap H_0^1$ with norm $\|D^2 z\|_2$.

Lemma 6.1 (Dirichlet Hessian estimate)**.** *For $z \in \mathcal{X}$, $\|D^2 z\|_2 \le \|\Delta z\|_2$.*

Proof. For a smooth zero-trace function, integrate $\operatorname{div}((\Delta z)\nabla z - D^2 z \nabla z) = (\Delta z)^2 - |D^2 z|^2$. On the boundary $\nabla z = (\partial_\nu z)\nu$ and the boundary expression is the sum of outward principal curvatures times $(\partial_\nu z)^2$, which is nonnegative by convexity. Approximation in $H^2 \cap H_0^1$ proves the assertion. On a box the same assertion follows from the orthogonal sine expansion, where the summed squared second derivatives equal the squared Laplacian norm. Standard Dirichlet solvability is an explicit domain hypothesis here, not asserted for arbitrary filled manifolds. □

Theorem 6.2 (Rank-one residual inverse)**.** *Let $e(x)$ be a measurable unit vector and $c_0 \le \chi(x) \le 1$, $c_0 > 0$. Define $A = I - (1-\chi)e \otimes e$ and*

$$Lz = (A + E) : D^2 z + b \cdot Dz + az, \quad \|E\|_{L^\infty(F)} \le \delta, \quad \|b\|_\infty \le b_0, \quad \|a\|_\infty \le a_0.$$

If $\|Dz\|_2 \le C_1 \|D^2 z\|_2$, $\|z\|_2 \le C_0 \|D^2 z\|_2$, and

$$\kappa = 1 - c_0 + \delta + b_0 C_1 + a_0 C_0 < 1, \tag{6.1}$$

then $L : \mathcal{X} \to L^2$ is an isomorphism and

$$\left\|D^2(z - z_c)\right\|_2 \le \frac{\|f - Lz_c\|_2}{1 - \kappa}, \qquad Lz = f. \tag{6.2}$$

Proof. Since the Frobenius norm of $(1-\chi)e \otimes e$ is $1 - \chi$, the map $Tz = \Delta_D^{-1}(f + (I - A - E) : D^2 z - b \cdot Dz - az)$ has Lipschitz constant at most κ in $\mathcal{X}$, by Lemma 6.1. This is a complete

normed space by Dirichlet solvability and Poincare inequalities. Banach's theorem gives existence and uniqueness. Applying the same estimate to the error gives $(1-\kappa)\left\|D^2(z-z_c)\right\|_2 \le \|f-Lz_c\|_2$. Surjectivity, injectivity, and bounded inverse all follow. No derivative of A, symmetry of the differential realization, or divergence conversion was used. □

For $E = b = a = 0$ the inverse constant is $1/c_0$, regardless of how rapidly the axis varies. This is an H^2 result only; it does not imply uniform $C^{1,\alpha}$ control for arbitrary measurable axes. In a smooth chart the covariant Hessian contributes connection drift; the metric deviation contributes E. The full linearization of (5.8) also contributes coefficient-dependent drift and zeroth-order terms, which must enter (6.1). Domain geometry, chart distortion and boundary lifts must be bounded explicitly. A local chart certificate cannot be silently glued into a global inverse.

Proposition 6.3 (Exact surrogate error)**.** *On $(0,1)^4$ let $\chi = (1+x_1)/2$, $e = e_1$, and $z = \prod_{i=1}^4 x_i(1-x_i)$. Then*

$$\operatorname{div}(A\nabla z) - A : D^2 z = \tfrac{1}{2}\partial_1 z \neq 0, \qquad \left\|\tfrac{1}{2}\partial_1 z\right\|_2^2 = \frac{1}{324000}.$$

Thus a residual computed with the symmetric divergence surrogate can vanish while the actual residual is nonzero.

Proof. The only nonconstant coefficient is $A_{11} = \chi$, so the product rule gives the displayed drift. Its norm factors into $\frac{1}{4}\int_0^1 (1-2x)^2 \mathrm{d}x$ times three copies of $\int_0^1 x^2(1-x)^2 \mathrm{d}x$, namely $\frac{1}{4}\cdot\frac{1}{3}\cdot 30^{-3}$. Choosing the right-hand side to be $\operatorname{div}(A\nabla z)$ proves the residual assertion. This example is not claimed to be an actual filled-system solution. □

7 A Natural-Growth Obstruction and a Morrey Budget

Theorem 7.1 (Quadratic gradients do not define the required energy-space map)**.** *For every $n > 2$ and every domain containing a ball, $z \mapsto |Dz|^2$ is not bounded on bounded subsets of H_0^1 with values in H^{-1}. Moreover, some $z \in H_0^1$ have $|Dz|^2 \notin H^{-1}$.*

Proof. Take compactly supported smooth ϕ, ψ in a ball, $\psi \ge 0$ and $\int |D\phi|^2\psi > 0$. Let $z_\epsilon(x) = \epsilon^{(2-n)/2}\phi(x/\epsilon)$ and $\psi_\epsilon(x) = \epsilon^{(2-n)/2}\psi(x/\epsilon)$ after centering the ball. Their gradient norms are constant, whereas

$$\langle |Dz_\epsilon|^2, \psi_\epsilon\rangle = \epsilon^{(2-n)/2}\int |D\phi|^2\psi \longrightarrow \infty.$$

For the second assertion choose a smooth cutoff $\zeta = 1$ near zero and $z = \zeta|x|^{-a}$ with $(n-2)/4 < a < (n-2)/2$. The gradient square is locally integrable and $z \in H_0^1$ because $2a < n-2$. Test against normalized nonnegative bumps on the annulus $\epsilon < |x| < 2\epsilon$. The pairing with $|Dz|^2$ is a positive constant times $\epsilon^{(n-2)/2-2a}$, which diverges. Their H_0^1 gradient norms stay fixed, contradicting membership in H^{-1}. □

This obstructs an unrestricted invocation of TR8's $H^{-1} \to H_0^1$ inverse for the coupled natural-growth equation, already in dimension four. It does not contradict the source's bounded-solution exponential test or the extra Morrey hypothesis. A strong-space repair on a fixed smooth bounded domain is $W^{2,p}$ with $p > n$, or $C^{2,\alpha}$: first derivatives are bounded and the nonlinear products lie in appropriate strong source spaces. Such a repair still needs a genuine inverse and tail controls.

Proposition 7.2 (Explicit local potential budget)**.** *If a nonnegative measure satisfies* $\mu(B_s(x)) \le M s^{n-2+\alpha}$ *for all* $s \le r$, $0 < \alpha < 1$, *then*

$$\int_{B_r(x)} |x-y|^{2-n}\,\mathrm{d}\mu(y) \le \frac{2^{n-2}M}{1-2^{-\alpha}} r^{\alpha}. \tag{7.1}$$

Proof. Split the punctured ball into annuli $2^{-j-1}r < |x-y| \le 2^{-j}r$. The kernel is at most $(2^{-j-1}r)^{2-n}$ and the measure is at most $M(2^{-j}r)^{n-2+\alpha}$. Sum the geometric series. The ball bound implies no atom at x, so nothing is omitted. □

If a scalar Green kernel is bounded by $C_G|x-y|^{2-n}$, multiply this local budget by C_G. Sources outside the ball require a separate tail estimate. This makes one source contribution auditable; it is not a proof of the coupled Holder theorem or a uniform-in-N Green constant.

8 Full Nonlinear Derivatives, Including Observation Dependence

All expressions in this section are on a fixed smooth background, with fixed K, N, ε, η and profile. Directions a, b are pairs $(f_a, Z_a), (f_b, Z_b)$, with homogeneous boundary traces. Covariant derivatives commute with these parameter variations. Set $q_a = \nabla f_a$, $H_a = \operatorname{Hess} f_a$. Write $h(t,q,Z) = t + (2k)^{-1}\log(1+l^2|q|^2) - Z$ and $J = 1 + pv/k$. Then

$$t_a = J^{-1}\left(Z_a - k^{-1}l^2 d\langle q, q_a\rangle\right), \tag{8.1}$$

$$\begin{aligned} h_{tt} &= k^{-1}(p'v + 2p^2 vd), \qquad h_{tq}[a] = 2k^{-1}pl^2d^2\langle q,a\rangle, \\ h_{qq}[a,b] &= k^{-1}l^2 d\langle a,b\rangle - 2k^{-1}l^4 d^2\langle q,a\rangle\langle q,b\rangle, \\ t_{ab} &= -J^{-1}\{h_{tt}t_a t_b + h_{tq}[q_a]t_b + h_{tq}[q_b]t_a + h_{qq}[q_a,q_b]\}. \end{aligned} \tag{8.2}$$

These are exact implicit derivatives, not a freeze of t, χ, or u.

8.1 A Finite Jet Specification With No Omitted Terms

The following rules define every first and mixed second variation in closed finite operations. For any scalar, vector, or tensor product/contraction AB,

$$(AB)_a = A_a B + AB_a, \quad (AB)_{ab} = A_{ab}B + A_a B_b + A_b B_a + AB_{ab}. \tag{8.3}$$

For a scalar composition $\varphi(A)$, use $\varphi(A)_a = \varphi'(A)A_a$ and $\varphi(A)_{ab} = \varphi''(A)A_aA_b + \varphi'(A)A_{ab}$. In particular $(A^{-1})_a = -A^{-2}A_a$ and $(A^{-1})_{ab} = 2A^{-3}A_aA_b - A^{-2}A_{ab}$. The required univariate derivatives are

$$l' = pl, \qquad l'' = (p' + p^2)l, \qquad p' = N^2\theta'(Nt), \qquad p'' = N^3\theta''(Nt), \tag{8.4}$$

$$m_0' = N\theta'(N(t+\varepsilon)), \qquad m_0'' = N^2\theta''(N(t+\varepsilon)). \tag{8.5}$$

Derivatives of powers, logarithms and exponentials have their usual exact formulas. Starting with affine jets $(f, Z, q, H, \mathrm{d}Z)$ (mixed second variations zero), evaluate (8.1)–(8.2), then the following ordered

directed graph:

$$
\begin{gathered}
p, l, D, d, c, w, v, \quad \gamma = (k+p)/(k+pv), \quad A = g^{-1} - \gamma w \otimes w, \quad A_d = g^{-1} - w \otimes w, \\
B = K + cH, \quad F = A : B, \quad u = e^{(k-1)t} l\sqrt{D}, \quad s_p = p + (k-1)/2, \\
T = \mathrm{d}t = (k\mathrm{d}Z - c^2 H(q,\cdot))/(k+pv), \quad Q_B = B + p(T \otimes w^\flat + w^\flat \otimes T), \quad P_B = Q_B - (\operatorname{tr} Q_B) g, \\
G = F + \gamma B(w,w) + 2s_p \langle w, T \rangle, \\
\mathcal{T} = \tfrac{1}{2}(|Q_B|^2 - (\operatorname{tr} Q_B)^2) + (k-1) P_B(w,T) + k s_p A_d(T,T) + FG.
\end{gathered} \tag{8.6}
$$

Here $A = A_\chi$; Q_B is a tensor, not electric charge. This invariant expression for $\mathcal{T}$ follows from source `alg:invariant-remainder` and `alg:flux-identities`; the term $\gamma B(w,w) = (1-\chi)j$ is smooth even at zero slope. The graph contains no division by $|q|$ and defines every derivative there without choosing a discontinuous unit axis.

For clarity, the final residual derivatives are explicit. Let $J_V = k\mathrm{d}Z + K(w,\cdot)$, $W_0 = \rho_0 + v\rho_1$, $S = \mathcal{T} + \eta c|q|^2$, and $\mathcal{R} = (F - \eta f, -\operatorname{div} V + u\Xi)$. Then

$$(\mathcal{R}_1)_a = A_a : B + A : B_a - \eta f_a, \tag{8.7}$$

$$(\mathcal{R}_1)_{ab} = A_{ab} : B + A_a : B_b + A_b : B_a + A : B_{ab}, \tag{8.8}$$

$$(\Xi)_a = \delta_0 S_a - C_N((m_0)_a W_0 + m_0 v_a \rho_1), \tag{8.9}$$

$$(\Xi)_{ab} = \delta_0 S_{ab} - C_N((m_0)_{ab} W_0 + (m_0)_a v_b \rho_1 + (m_0)_b v_a \rho_1 + m_0 v_{ab} \rho_1), \tag{8.10}$$

$$V_a = u_a A J_V + u A_a J_V + u A (J_V)_a, \tag{8.11}$$

$$\begin{aligned} V_{ab} = {} & u_{ab} A J_V + u_a A_b J_V + u_b A_a J_V + u A_{ab} J_V \\ & + u_a A (J_V)_b + u_b A (J_V)_a + u A_a (J_V)_b + u A_b (J_V)_a + u A (J_V)_{ab}, \end{aligned} \tag{8.12}$$

$$(\mathcal{R}_2)_a = -\operatorname{div} V_a + u_a \Xi + u \Xi_a, \tag{8.13}$$

$$(\mathcal{R}_2)_{ab} = -\operatorname{div} V_{ab} + u_{ab} \Xi + u_a \Xi_b + u_b \Xi_a + u \Xi_{ab}. \tag{8.14}$$

Here $(J_V)_a = k\mathrm{d}Z_a + K(w_a,\cdot)$ and $(J_V)_{ab} = K(w_{ab},\cdot)$. For example $S_a = \mathcal{T}_a + \eta(c_a|q|^2 + 2c\langle q, q_a\rangle)$ and

$$S_{ab} = \mathcal{T}_{ab} + \eta(c_{ab}|q|^2 + 2c_a\langle q, q_b\rangle + 2c_b\langle q, q_a\rangle + 2c\langle q_a, q_b\rangle).$$

The jets of $\mathcal{T}$ are *fully specified* by the finite graph (8.6) and rules (8.3); no unspecified kernel or observation derivative is suppressed. Induction on this graph proves that the displayed operations are the first and second Frechet derivatives. The implicit-function theorem proves the initial t jets; product and composition rules prove each subsequent node. In the weak second component, replace $-\operatorname{div} V_a$ by $\int V_a \cdot \mathrm{d}\varphi$ and add $\int (u\Xi)_a \varphi$; the sign is fixed by (5.5).

8.2 Feasible Local Certification

On a fixed smooth compact filled truncation, take $\mathcal{X} = (C^{2,\alpha} \cap C_0)^2$ and $\mathcal{Y} = (C^{0,\alpha})^2$. On bounded sets with t in a compact interval the residual is $C^2 : \mathcal{X} \to \mathcal{Y}$ by the preceding graph. The principal diagonal blocks are multiples of $A_\chi :$ Hess and $-kuA_\chi :$ Hess; lower and off-diagonal terms remain. Ellipticity alone does not imply the full coupled Dirichlet derivative is invertible.

Theorem 8.1 (Feasible residual ball)**.** *Let $x_c = (f_c, Z_c)$, and let $B : \mathcal{Y} \to \mathcal{X}$ be a bounded isomorphism. On the closed radius-r ball suppose*

$$a = \|B\mathcal{R}(x_c)\|, \qquad \sup \|I - BD\mathcal{R}(x)\| \le q < 1, \qquad a + qr \le r.$$

Then the ball has a unique zero of $\mathcal{R}$. If $t(x_c) \ge -\varepsilon + \sigma$ and $L_t r < \sigma$, where $\|t(x) - t(x_c)\|_\infty \le L_t \|x - x_c\|_{\mathcal{X}}$ on the ball, that zero remains above the source floor.

Proof. The map $T(x) = x - B\mathcal{R}(x)$ is q-Lipschitz by integration of its derivative on segments. Its distance from x_c is at most $a + qr$, so it maps the complete ball into itself. The contraction theorem gives a unique fixed point; injectivity of B makes that point a zero. The last inequality gives $t > -\varepsilon$ everywhere. □

A computable sufficient bound is $q \le q_0 + \|B\| Lr$, where $q_0 = \|I - BD\mathcal{R}(x_c)\|$ and L bounds the second derivative. Thus one tests $a + (q_0 - 1)r + \|B\| Lr^2 \le 0$, as in the actual TR1 theorem, *together with* the feasibility margin. Formula (8.1) bounds L_t using embedding constants. Similar strict margins can be imposed on ellipticity and selected regular levels. A finite matrix inverse, without a validated complement/tail inverse and norm transfer, does not meet the theorem. No such full inverse for a prepared source geometry is produced in this report.

9 Charged Flux Conversion: Exact Energy-Space Transfer

Fix $0 < s < 1$, $y_0 = \log(1 - s^2)/(k - 1)$. Source profiles satisfy $L > 0$, $L(0) = 1$, $L' \ge 0$, and $H_0 = 0$ below y_0 with $H_0' \ge 0$ supported compactly in $(y_0, 0)$. They obey

$$|H_0'| \le k\sqrt{(k-1)(k-1+2L'/L)}\, e^{(k-1)y} L, \quad ks - \delta < H_0(0) \le ks. \tag{9.1}$$

Define $B_0(0) = 0$, $B_0' = e^{(k-1)y} L$, $Y = B_0(y)$ and $\mathfrak{h}(Y) = H_0(B_0^{-1}(Y))$, extended by zero below $Y_0 = B_0(y_0)$ and by its constant upper value. This gives a smooth bounded globally Lipschitz function for the fixed profile. Keep $\alpha = \iota_E \operatorname{vol}_g$ and Q for the original data. For conversion only, choose $s_Q = \operatorname{sgn}(Q)$ when $Q \ne 0$ and either $s_Q \in \{-1, 1\}$ when $Q = 0$. Extend $s_Q\alpha$ into the filling as $\widetilde{\alpha}$, allowing filling sources, exactly as in source `flux:extended-fields`. With $z = t + \varepsilon$ and $g_b = e^{2z}(g + l^2 \mathrm{d}f^2)$, define

$$\iota_{E_{\mathrm{ext}}} \operatorname{vol}_g = \widetilde{\alpha}, \qquad \iota_{E_b} \operatorname{vol}_{g_b} = \widetilde{\alpha}. \tag{9.2}$$

The source equation with these sign-normalized conversion fields is

$$k \operatorname{div}_g(A \nabla_g Y) = \operatorname{div}_g(\mathfrak{h}(Y) E_{\mathrm{ext}}), \quad A = e^{(n-2)z}\sqrt{D}\, A_d. \tag{9.3}$$

Its eigenvalues in a g-orthonormal frame are $e^{(n-2)z}\sqrt{D}$ transversely and $e^{(n-2)z}/\sqrt{D}$ axially. The volume Jacobian $J_b = e^{nz}\sqrt{D}$ satisfies $J_b E_b = E_{\mathrm{ext}}$. Their exterior flux is $\omega s_Q Q = \omega|Q|$, not the unflipped original flux ωQ. This actual SPD divergence operator, unlike (5.8), fits TR8's energy inverse.

Theorem 9.1 (Fixed-domain conversion certificate)**.** *Let U be a fixed bounded smooth Riemannian domain, A measurable symmetric with $\lambda I \le A \le \Lambda I$, and $\|v\|_{2n/(n-2)} \le S \|\nabla v\|_2$ for $v \in H_0^1(U)$. Let $E \in L^n(U)$, and let $\mathfrak{h}$ be bounded and globally Lipschitz with constant L_h. If*

$$\kappa_f = S L_h \|E\|_n / (k\lambda) < 1, \tag{9.4}$$

then (9.3) with $Y = 0$ on ∂U has a unique weak solution. If

$$\mathfrak{r}_c(\varphi) = k \int A \nabla Y_c \cdot \nabla \varphi - \int \mathfrak{h}(Y_c) E \cdot \nabla \varphi,$$

then

$$\|\nabla(Y - Y_c)\|_2 \le \frac{\|\mathfrak{r}_c\|_{H^{-1}}}{k\lambda - S L_h \|E\|_n}, \tag{9.5}$$

where H^{-1} is dual to the gradient norm. If $\mathfrak{h} = 0$ for $Y \le Y_0 < 0$, the solution also satisfies $Y \ge Y_0$ almost everywhere.

Proof. Given $W \in H^1_0$, use the coercive form $k \int A\nabla Y \cdot \nabla\varphi$ and right side $\int \mathfrak{h}(W)E \cdot \nabla\varphi$ to define $T(W)$ by Lax–Milgram. Boundedness of $\mathfrak{h}$ and finite volume ensure $\mathfrak{h}(W)E \in L^2$. Holder and Sobolev give

$$\|(\mathfrak{h}(W_1) - \mathfrak{h}(W_2))E\|_2 \le L_h \|E\|_n S \|\nabla(W_1 - W_2)\|_2 .$$

Testing the difference equation proves that T contracts by κ_f. Subtract the equation at Y from the residual at Y_c, test with $Y - Y_c$, and use the same bound to prove (9.5). Finally test the weak solution with $(Y_0 - Y)_+ \in H^1_0$. The right side vanishes on its support, and the left side is $-k \int_{Y<Y_0} A\nabla Y \cdot \nabla Y$, so the truncation is constant zero by its boundary trace. □

No condition $\operatorname{div} E = 0$ was used here, so the theorem respects the source-bearing filling. For prescribed nonzero trace use a lift b and iterate on $Y = b + H^1_0$; the difference estimate is unchanged and the lift residual is mandatory. Coefficient perturbation δA and flux perturbation δE contribute at a candidate at most $k \|\delta A \nabla Y_c\|_2 + \|\mathfrak{h}(Y_c)\delta E\|_2$ to the dual residual, plus the profile perturbation term. This provides a practical error budget with the correct operator.

Condition (9.4) is restrictive. The source does not assume it, and establishes general existence by a priori estimates and degree. Even though $E = O(r^{-k})$ makes the far-end L^n norm finite, no uniform lower ellipticity bound as $N \to \infty$ has been proved here. Failure of the contraction test is not failure of the source equation.

9.1 Dimension-Specific Budget

The late trace construction spends the charged gradient margin with coefficient

$$\chi_k = \frac{2k^2}{k^2 - 1} + \frac{k}{2(k-1)}, \qquad d_k = \frac{k(k+1)}{2} - \chi_k. \tag{9.6}$$

For $k \ge 3$, the two summands decrease and hence $\chi_k \le \chi_3 = 3$, while $k(k+1)/2 \ge 6$; therefore $d_k \ge 3$. Also $3 - \chi_k = (k-3)(k+2)/(2(k-1)(k+1))$. For $k = 2$, $\chi_2 = 11/3$ and $d_2 = -2/3$: the same budget cannot be copied unchanged to dimension three. This is a concrete dimensional reason for this report's scope, not merely a relabeling of a low-dimensional argument.

9.2 The Charged-to-Neutral Handoff, With All Hypotheses Retained

The preceding local certificates do not themselves imply a Penrose bound. The source's remaining handoff in `flux:prepared-bound` is as follows. On the source-free original exterior, using the sign-normalized form in (9.2), the conversion equation has the exact asymptotic flux

$$\lim_{r\to\infty} \int_{S_r} \partial_\nu y = \frac{H_0(0)}{k}\omega q_e, \qquad \mathcal{E}(\gamma) = \mathcal{E}(g_b) - \frac{H_0(0)}{k} q_e, \quad \gamma = e^{2y} g_b.$$

The lower bound $y \ge y_0$ and $g_b \ge g$ give $\gamma \ge e^{2y_0} g$. Curvature is nonnegative on the designated low-height original region, with an extra positive band margin. This is a regional statement, not a global assertion on the filling.

Choose a regular height $c_h \in (b_3/2, b_3)$ after the two solves, and retain the closed end component of $\{|h| < c_h\}$. Height separation keeps it outside the filling. Its full boundary is nonempty, smooth, compact and contains every component of that boundary. The source adds the compactly supported pure-trace tensor $K_{\text{new}} = -\mathfrak{a}\gamma$, with $\mathfrak{a} = e^{-(z+b(y))}\mathcal{H}(|h|)$. For a fixed cutoff χ_h, the ramp $\mathcal{H} = \tan((\arctan M)\chi_h)$ satisfies $|\mathcal{H}'|/(1+\mathcal{H}^2) \le (\pi/2)\|\chi_h'\|_\infty$, independent of finite M. Since $\eta/l \le \ell^{1/2}$, sufficiently large N

fixes the DEC budget before M is chosen. A late finite M then makes every component strictly future trapped. This order prevents the plateau choice from invalidating either PDE solve.

The new exterior is connected and complete with boundary, has one end, all-order r^{2-n} metric decay, integrable neutral densities, zero ADM momentum, and the required future boundary sign. Every full cut there is a full cut in the original exterior, including contact portions. Hence its infimum satisfies $a_N \geq e^{ky_0} a_* > 0$ without assuming that a minimizing cut is smooth or attained. The positivity of its energy e_N is *not* silently supplied by (2.3). If $e_N \leq 0$, the source's conformal end increase preserves these properties, does not decrease the area, and changes energy to $e_N + 2\lambda$. Choosing $\lambda = (e - e_N)/2$ produces arbitrarily small positive e. Applying the neutral input to those timelike data would force $e \geq \frac{1}{2} e^{(k-1)y_0} X_* > 0$, a contradiction. Only then is the input applied directly to the unmodified auxiliary data.

Combining the actual mass decrease, the filled mass-cost estimate and the retained area gives

$$e^{(k-1)\varepsilon}(\mathcal{E}_* + o_N(1)) - \frac{H_0(0)}{k} q_e \geq \frac{1 - s^2}{2} X_*. \tag{9.7}$$

The ordered limits in Section 5.2 yield $\mathcal{E}_* \geq s q_e + (1 - s^2) X_*/2$. Rest preparation has $\mathcal{E}_{*,j} \to m$ and $\liminf a_{*,j} \geq a$, so passing back to the original data preserves the lower bound. Optimization is the elementary branch step in Proposition 10.3. This audit displays where each global source obligation enters; it does not independently prove the preparation, regional curvature, degree, exhaustion, or neutral theorem.

10 Charge Forms, Gauge, Magnetic Fields, and Branch Guards

10.1 What Must Remain Fixed

On the original exterior $\alpha = \iota_E \operatorname{vol}_g$ is a closed $(n-1)$-form. Under $g_s = e^{2s} g$ with α fixed, $E_s = e^{-ns} E$; under the graph metric the exact norm identity is

$$|\alpha|^2_{g + l^2 \mathrm{d}f^2} = d|E|^2 + \langle w, E \rangle^2 = |E|^2 - |w \wedge E|^2. \tag{10.1}$$

Conformal form norms scale by e^{-2ks}, not the vector scaling guessed by keeping coordinate components of E fixed. The source's charge-adjusted inequality is retained only on the low-height original region; no charged DEC is asserted throughout the filling.

Proposition 10.1 (Filling obstruction)**.** *Suppose a compact oriented filling W has the full original boundary as its only boundary and a smooth closed extension $\widetilde{\alpha}$ agreeing with α there. Then the total electric flux through that boundary is zero. Consequently a nonzero total charge cannot admit such a source-free filling.*

Proof. Stokes' theorem gives $\int_{\partial W} \widetilde{\alpha} = \int_W \mathrm{d}\widetilde{\alpha} = 0$. On the source-free exterior the total oriented boundary flux equals the flux at infinity, up to the consistent boundary orientation. Thus nonzero Q contradicts the extension. This concerns total flux; opposite component fluxes can cancel and are not excluded by this argument. □

The source correctly cuts off a collar extension *inside* the added region. Any moment model must carry this source and the flux interface, not silently impose global divergence zero.

10.2 Exact Magnetic Graph Test

In the static exterior of RNT,

$$\mathbf{G} = -f_r \mathrm{d}T^2 + f_r^{-1}\mathrm{d}r^2 + r^2\sigma, \quad f_r = 1 - 2Mr^{-(n-2)} + Q^2 r^{-2n+4}, \quad \mathbf{F} = c_n Q r^{-k}\mathrm{d}T \wedge \mathrm{d}r.$$

Here f_r denotes the radial lapse function, not a derivative of the filled unknown.

Proposition 10.2 (Purely electric graph obstruction)**.** *Let a smooth spacelike graph in this static exterior be $T = \tau(r, \vartheta)$, over full connected spherical sections, and suppose $Q \neq 0$. Its induced magnetic two-form vanishes if and only if τ is independent of ϑ. Relative to the reference spatial metric $h = f_r^{-1}\mathrm{d}r^2 + r^2\sigma$,*

$$|\iota^*\mathbf{F}|_h^2 = c_n^2 Q^2 f_r r^{-2k-2} |\mathrm{d}_{S^k}\tau|_\sigma^2. \tag{10.2}$$

Proof. Pullback gives $\iota^*\mathbf{F} = c_n Q r^{-k}\mathrm{d}\tau \wedge \mathrm{d}r = c_n Q r^{-k}\mathrm{d}_{S^k}\tau \wedge \mathrm{d}r$. Since radial and angular covectors are h-orthogonal, their wedge norm squared is the product of their squared norms, f_r and $r^{-2}|\mathrm{d}_{S^k}\tau|_\sigma^2$. This proves (10.2). Positivity of f_r in the static exterior and $Q \neq 0$ imply angular derivative zero. A smooth function with zero differential on a connected sphere is constant there. The converse is immediate. □

The norm in (10.2) is explicitly the reference norm, not the induced graph norm. Vanishing is metric independent, so the admissibility conclusion is unchanged. The theorem is restricted to a graph class and does not prove global graph existence or horizon regularity. At $Q = 0$ it gives no restriction. In particular, a generic angularly tilted or boosted graph cannot be declared a purely electric equality example merely by recognizing the ambient RNT metric. The source's radial compactly supported tilts pass this test and can have nonzero K.

Gauge choices are also constrained. The source gets $\mathrm{d}p = (k-1)uE^\flat$, $p_\infty = 0$, after testing all compactly supported closed electric variations. Testing only exact forms loses cohomological information. The stationary one-form $W^{-1}X^\flat$ is first shown closed across the relevant region; its global primitive enters only after the exterior topology is identified. Imposing an arbitrary global time gauge beforehand discards a proof obligation.

10.3 Branch-Safe Error Propagation

Proposition 10.3 (Exact optimization with additive budgets)**.** *Suppose $X > 0$, $q_e \geq 0$ and for every $0 < s < 1$,*

$$m \geq sq_e + (1-s^2)X/2 - e_0 - e_1 s - e_2(1-s^2), \quad e_i \geq 0.$$

Set $X' = X - 2e_2 > 0$, $q' = q_e - e_1$. Then

$$m + e_0 \geq \begin{cases} X'/2, & q' \leq 0, \\ (X' + (q')^2/X')/2, & 0 < q' < X', \\ q', & q' \geq X'. \end{cases} \tag{10.3}$$

Proof. The right side before optimization is the concave quadratic $sq' + (1-s^2)X'/2$. Its derivative is $q' - sX'$. For $q' \leq 0$ the supremum is the $s \downarrow 0$ endpoint; for $0 < q' < X'$ it is attained at $s = q'/X'$; for $q' \geq X'$ it is the $s \uparrow 1$ endpoint. Continuity justifies using open-interval endpoints as limits. □

With zero errors this is exactly the branch logic of `prep:prepared-reduction`. A rigorous interval implementation may use the polynomial branch only after proving $X_{\mathrm{lower}} > q_{e,\mathrm{upper}}$. For example $m = q_e = 1$, $X = 1/2$ satisfies (2.4) but fails $m \geq (X + q_e^2/X)/2 = 5/4$. Hence a branch guard is mathematically necessary. The budget theorem does not furnish the required uniform e_i across all s or source limits; those must be proved separately.

11 Exact Moments and Bounded Computation

On a fixed strong trial space $V_h \subset H^2 \cap H_0^1$ with basis ϕ_i, define the *actual* strong residual Gram matrix $G_{ij} = \int_U (L\phi_i)(L\phi_j)$. For a residual r in the span of $L\phi_i$, its squared L^2 norm is the exact quadratic form of its coefficient vector and G. A positive smallest eigenvalue controls only this finite span; it is not a bound on a missing complement. If $r = r_h + r_\perp$ is an orthogonal split with a separately proved $\|r_\perp\| \le \epsilon_\perp$, then $\|r\| \le (\|r_h\|^2 + \epsilon_\perp^2)^{1/2}$. The proof is the Pythagorean identity. Feeding that *full* bound into Theorem 6.2 is legitimate. Feeding a finite residual alone is not.

For the true divergence equation (9.3), weak assembly instead uses $k \int A\nabla\phi_i \cdot \nabla\phi_j$ and $\int \mathfrak{h}(Y_h)E \cdot \nabla\phi_j$. For the nondivergence operator, integration by parts differentiates the coefficient times the test and introduces its real boundary terms. TR1's `eq:weak` and `eq:boundaryweak` explicitly require this. Exactness of polynomial moments removes quadrature error, not modeling error, boundary error, nonlinear remainder, or infinite-dimensional tail error.

11.1 Reproducible Test Contract

The accompanying `verify_M511.py`, run with `py -3.10`, uses SymPy exact rational algebra. Its receipt is `_audit/M511-computation-receipt.json`. Tests cover:

1. the rank-one eigenvalue and comparison identities, and the exact nondivergence-surrogate defect in Proposition 6.3;
2. equality of the source block remainder and the smooth invariant remainder in (8.6), checked symbolically in a four-dimensional diagonal/transverse representative with all block variables retained;
3. exact first and second implicit derivatives, checked against direct differentiation of h for the locally constant-p profile $l = e^{pt}$; the variable-p and p' terms are proved analytically above, not inferred from that restricted test;
4. the all-k slack factorization in (9.6), the $n = 4$ failure exponent in Theorem 7.1, and rational branch examples;
5. a synthetic residual-ball budget $a = 1/100$, $q_0 = 1/4$, $\|B\|\, L = 1$, $r = 1/20$, with $q \le 3/10$ and $a + qr = 1/40 < r$; the floor budget $L_t = 2$, $\sigma = 1/5$ is strictly satisfied;
6. a synthetic conversion budget $k = 3$, $\lambda = 1/2$, $SL_h \|E\|_n \le 1/4$, residual at most $1/100$, giving $\kappa_f \le 1/6$ and error at most $1/125$;
7. the negative-charge regression: for $Q = -1$, $k = 3$, $H_0(0) = 3/2$, an unflipped extension would give $\Delta\mathcal{E} = -H_0(0)Q/k = +1/2$, a mass increase. The normalized form gives $\Delta\mathcal{E} = -H_0(0)|Q|/k = -1/2$, the correct decrease. These are exact flux/mass arithmetic, not a solved geometric example.

The exact verification run passed all 42 author checks with no failures, using Python 3.10 and SymPy 1.11.1, including source-label and text-integrity checks. The two budget examples are *conditional arithmetic examples*, not measured constants from a prepared initial data set. The independent review's eight in-memory checks have a separate scope and are not added to this author count. No mesh of the filled source geometry, no actual source-profile solution, no certified coupled inverse, no enclosure of an ADM mass, and no singular area envelope is computed. Exact identities and rational inequalities have zero roundoff error; their application to geometry still requires analytic assumptions. Evidence receipts state this scope rather than labeling toy tests a global proof.

12 Rigidity Interfaces That Remain Outside the Certificate

The high-dimensional modules contain substantial independent obligations. Minimizing enclosures may have a singular set of dimension at most $n - 8$. Their area multiplier is a measure, not a sampled

smooth function. The source uses the actual gravitational adjoint, a positive normal-normal singular Hessian term, smoothing commutators, reciprocal Jacobi estimates, capacity bounds, and BV locality to eliminate interior multiplier mass. Neither an SPD matrix nor a finite moment fit proves the required tensor positivity or absence of diffuse mass.

After the source produces stationary data, $W = u^2 - |X|^2$ can initially vanish. The alignment, twist and positivity arguments must cross this zero set before defining the static base everywhere. For nonzero charge the source obtains $W = (1 - p/p_b)(1 - p_b p)$ with $0 < |p_b| < 1$, a bounded potential range, and then a vacuum transform. For zero charge it uses a separate complete circle-bolt argument. These are not consequences of the scalar Y contraction.

At positive boundary lapse the attached static base uses the new coordinate $\lambda = \sqrt{W}$; the inverse of the squaring correspondence is not smooth in the original boundary structure. At zero lapse the original collar is retained. The conformal double in module 09 first has a rough compactification, which must be regularized and shown flat through the source's zero-mass argument. That argument in turn uses its completed numerical theorem at $Q = 0$. The global recovery of the original slice and its future horizon section therefore remains conditional on the full chain, including (2.3).

The strict compact-magnetic extension in the source concerns arbitrary smooth compactly supported closed two-forms under its specified electromagnetic energy convention, strict DEC margin, strict trapping and already future-timelike assumptions. It is not a general magnetic equality theorem, not a treatment of arbitrary electromagnetic boosts, and not an extremal endpoint result. Our graph obstruction is compatible with, and narrower than, that extension.

13 Literature Position, Improvements, and Future Work

Public literature was checked on October 8, 2026 with public mathematical query terms only. Smears–Suli's nondivergence Cordes analysis [4] supports taking operator form seriously; our rank-one estimate is a direct small-perturbation consequence of the Dirichlet Hessian estimate, not a claimed new general elliptic theory. Kang–Kim [5] distinguish scalar Green bounds from systems requiring additional hypotheses; our local Morrey budget assumes the requisite scalar bound explicitly. Khuri–Weinstein–Yamada [6] is Riemannian charged work, not an unrestricted higher-dimensional spacetime input. Gibbons–Ida–Shiromizu [7] concerns static higher-dimensional charged uniqueness, not a replacement for the original-data stationary reduction. The current Bi–Zhu preprint [8], revised September 15, 2026, concerns the Riemannian Penrose inequality in all dimensions; it is not the neutral spacetime theorem (2.3). These references delimit context and do not certify the assigned source. No exhaustive novelty search or priority claim is made.

Concrete improvements. The report supplies a zero-slope-safe derivative graph, keeps the first operator in nondivergence form, gives a full residual denominator on bounded convex patches, proves an energy-space obstruction rather than assuming natural growth is harmless, and gives a genuine energy inverse only for the conversion equation. The electric form, filling source, magnetic pullback, and branch guard are explicit constraints. Exact arithmetic makes the proposed tests reproducible.

Advantages. The local theorems have complete short proofs and identifiable hypotheses. They distinguish strong and weak residuals and preserve boundary conditions. The $k \geq 3$ slack and source exponents are dimension specific. The derivative graph supports interval differentiation without unstable divisions by slope. The conversion certificate accommodates nonzero filling divergence without a false source-free extension.

Costs and limitations. The strong inverse is local and assumes appropriate domain solvability; its drift allowance may be too small. The conversion smallness criterion may fail for the actual profile and field. The coupled residual ball requires an unconstructed full-space inverse. Constants can deteriorate with N, and no computation here controls exhaustion, ADM tails, full-cut area, source profile smoothing, singular multipliers, or rigidity. The derivative formulas are for a fixed background; metric, shape, or charge variations need their additional derivatives and gauge constraints.

Priority future work. First choose one explicit prepared nontrivial data set and fix N, R, all profiles and boundary lifts. Build an H^2-conforming or validated nondivergence discretization and a complement inverse; enclose the jets and floor margin on an actual residual ball. Independently certify the flux equation with its real coefficient field, using degree-compatible verification if contraction fails. Next establish exterior tail and flux enclosures uniform in R at that fixed N. Only after those steps consider the much harder N-dependent mass-cost budget. A separate project would be required for a rigorous singular-multiplier or neutral-input verification. None is silently delegated to a representation method.

14 Contribution Assessment and Delivery Evidence

C2/5, independently of L4. Literal mathematical evidence is in Sections 6, 7, 8, 9, and 10. The contribution is a source-specific set of restricted certificates and obstructions plus a complete implementable derivative specification. Generic restatement of contraction mapping, Lax–Milgram, or the source theorem is not scored as a new global result. This report does not establish literature-level novelty of those local tools and does not merit C4/C5 on the basis of its breadth.

The assessment sidecar uses the schema of `_audit/contribution-assessment-M442.json`, with literal section labels and SHA-256 of this report's final LaTeX source. The frozen manifest records report, test and evidence hashes. Author native open/compile dispatch attempts failed because the tools were not exposed; no author native backend ran (`_audit/M511-native-attempts.json`). The parent separately reports an actual native open queued and a native compile platform-directory failure; neither is claimed as an author backend call or native compilation success. The original two minor formatting review rounds are preserved. The essential charge-sign correction uses a separate `M511-charge-correction` physical-export tag at round 2, not a third minor-format iteration. Build and visual receipts distinguish inspection from compilation and retain the independent review's scope. No global theorem certification is inferred from export.

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